DSA

LAB 7

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Q1:

Code:

```
#include<iostream>
#include<vector>
#include<stack>
using namespace std;
int partition(vector<int>& arr, int st, int end) {
int idx = st - 1, pivot = arr[end];
for (int j=st; j<end; j++) {
if (arr[j]<=pivot) {
idx++;
swap(arr[j], arr[idx]);
}}
idx++;
swap(arr[end], arr[idx]);
return idx;}
void quickSortIterative(vector<int>& arr, int st, int end) {
stack<int> s;
s.push(st);
s.push(end);
while(!s.empty()){
end =s.top();
s.pop();
st=s.top();
s.pop();
if(st<end){
int middle=partition(arr, st, end);
if(middle+1< end){
s.push(middle+1);
s.push(end);
}
```

```cpp
if(st<(middle-1)){
s.push(st);
s.push(middle-1);
}}}}
int main() {
vector<int> arr = {12, 31, 35, 8, 32, 17};
cout <<"\nbefore applying quick sort"<< endl;
for(int val : arr) {
cout<< val<<" ";
}
quickSortIterative(arr, 0, arr.size() - 1);
cout<< "\nafter applying quick sort non recursively using stack"<< endl;
for(int val:arr) {
cout <<val<< " ";
}
cout << endl;
}
```

Output:

```
"F:\3rd semistor\dsa\labs\lab

before applying quick sort
12 31 35 8 32 17
after applying quick sort non recursively using stack
8 12 17 31 32 35

Process returned 0 (0x0)   execution time : 0.318 s
Press any key to continue.
```

Q2:

Code:

```
#include<iostream>
using namespace std;
void merge(int arr[],int left,int mid,int right){
int n1=mid-left+1;
int n2=right-mid;
int L[n1];
int R[n2];
for(int i=0;i<n1;i++){
L[i]=arr[left+i];
}
for(int j=0;j<n2;j++){
R[j]=arr[mid+ j+ 1];
}
int i=0;
int j=0;
int k=left;
while(i<n1 && j<n2){
if(L[i]<=R[j]){
arr[k]=L[i];
i++;
}
else{
arr[k]=R[j];
j++;
}
k++;
}
while(i<n1){
arr[k]=L[i];
```

```cpp
i++;
k++;
}
while(j<n2){
arr[k]=R[j];
j++;
k++;}}
void mergesort(int arr[],int n){
for(int size=1; size<n; size*=2){
for(int left=0; left<n-1; left+=2*size){
int mid=min(left+size-1,n-1);
int right=min(left+2*size-1,n-1);
merge(arr,left,mid,right);
}}}
void printArray(int arr[], int size)  {
for (int i = 0; i < size; i++)
cout << arr[i] << " ";
cout << endl;
}
int main(){
int arr[] = {15, 12, 10, 4, 9, 1};
int n=sizeof(arr)/sizeof(arr[0]);
cout << "Unsorted array: ";
printArray(arr, n);
mergesort(arr,n);
cout << "Sorted array: ";
printArray(arr,n);
}
```

Output:

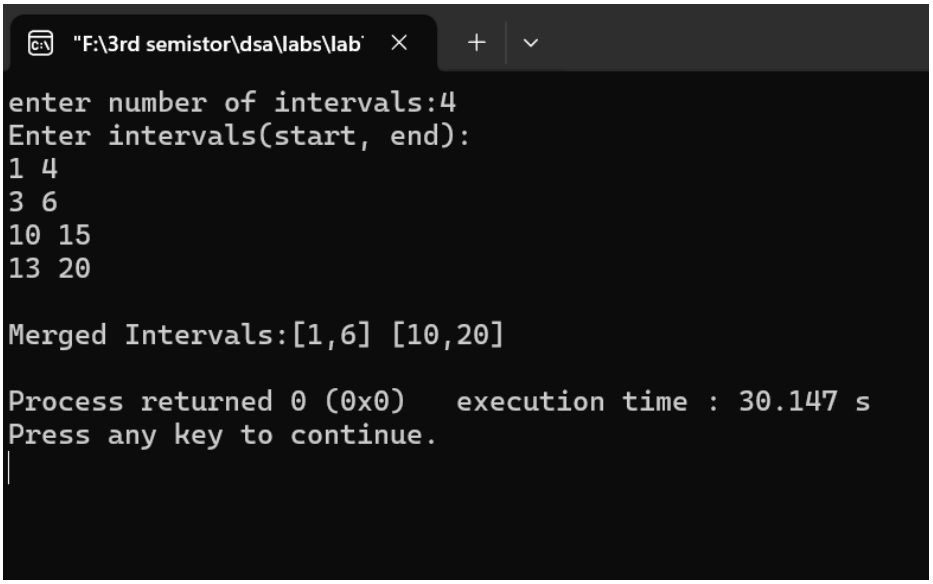
```
"F:\3rd semistor\dsa\labs\lab

Unsorted array: 15 12 10 4 9 1
Sorted array: 1 4 9 10 12 15

Process returned 0 (0x0)   execution time : 0.094 s
Press any key to continue.
```

Q3:

Code:

```cpp
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;
int main(){
int n;
cout<<"enter number of intervals:";
cin >> n;
vector<vector<int>> intervals(n, vector<int>(2));
cout<<"Enter intervals(start, end):"<<endl;
for(int i = 0; i < n; i++){
cin>>intervals[i][0]>>intervals[i][1];
}
sort(intervals.begin(), intervals.end());
vector<vector<int>> merge;
for (int i = 0; i < n; i++){
int start = intervals[i][0];
int end = intervals[i][1];
if (merge.empty()){
merge.push_back({start, end});
}else if(start <= merge.back()[1])  {
if(end > merge.back()[1]){
merge.back()[1] = end;
}}
else{
merge.push_back({start, end});
}
}
cout<<"\nMerged Intervals:";
for(int j = 0; j < merge.size(); j++)  {
cout<<"["<<merge[j][0]<<","<<merge[j][1]<<"] ";
}
cout<<endl;
return 0;
}
```

Output:




```
enter number of intervals:4
Enter intervals(start, end):
1 4
3 6
10 15
13 20

Merged Intervals:[1,6] [10,20]

Process returned 0 (0x0)   execution time : 30.147 s
Press any key to continue.
```